arXiv:2406.07756v4 [stat.ME] 9 Oct 2026

Exploring Exchangeability within Permutation Inference

Implications for Statistics and Data Science Educators

Johanna Hardin Lauren Quesada Julie Ye
Nicholas J. Horton

2026-09-26

Permutation tests are a powerful and flexible approach to inference via resampling. As computational methods become more ubiquitous in the statistics curriculum, use of permutation tests has become more tractable. At the heart of the permutation approach is the *exchangeability condition*, which determines the appropriate null sampling distribution. We explore the exchangeability condition in the context of permutation tests, including settings where the exchangeability condition is not tenable. Our examples include two-sample comparisons of means, clustered designs, multiple linear regression, and two-sample comparisons of correlation. The different exchangeability challenges in each example allow further exploration of the definition and importance of the exchangeability requirement for permutation testing. We believe that understanding the exchangeability condition and how it relates to the null hypothesis gives students the skills to work effectively with permutation tests in more complicated modeling settings and to further their understanding of statistical inference. We close with pedagogical recommendations for instructors who wish to incorporate permutation inference into their curriculum as a means to increase student understanding of resampling-based inference.

1 Introduction

Statistical inference attempts to infer properties of population parameters from sample statistics. Many of the models and analysis methods taught in a standard undergraduate statistics curriculum rely on asymptotic normal theory, with the theoretical underpinnings driven by the Central Limit Theorem. However, permutation tests, a form of resampling-based inference,[1] are becoming increasingly popular because they provide a flexible approach for a wide scope of problems, particularly those with small sample sizes.

Unlike methods based on the Central Limit Theorem, permutation tests do not generally require distributional (parametric) or sample size conditions. Instead, they require **exchangeability**, an idea which comprises much of the substance of our paper and which will be introduced in Section 1.1. Permutation tests were among the first inferential tests conceived, and they were used widely (Fisher 1935) in the context of categorical data and two-sample comparisons.

Since their introduction, permutation tests have been extended to address many different modeling contexts and are often presented to support a comprehensive understanding of sampling distributions and normal theory methods. As tools for testing that take full advantage of the computer, permutation tests have become part of what we think of as "computational statistics." The undergraduate statistics curriculum has historically not had space for extended explication of permutation methods; however, many modern introductory textbooks do intro-

[1] See Hesterberg (2015) for an excellent overview of bootstrapping, a form of resampling with replacement that works well with sufficiently large sample sizes.

duce the basics of permutation tests as part of the core content of the course (Ismay and Kim 2020; Chance and Rossman 2021; Baumer et al. 2024; Çetinkaya-Rundel and Hardin 2023; Lock et al. 2025). And while permutation tests have been shown to reinforce understanding of sampling distributions (Garfield et al. 2012; Taylor and Doehler 2015) and other inferential concepts, the *theory* that underlies permutations tests is typically missing from the conversation. In our experience, students at the undergraduate level are rarely given any guidance on the technical conditions required for permutation tests. To be fair, the presentation of permutation tests in undergraduate texts is almost always given under the *iid* (independent and identically distributed) setting, which is sufficient for permutation tests conditions. However, the *iid* setting excludes more complex scenarios for which permutation tests can be quite useful.

In what follows, we describe and detail the idea of exchangeability, which is the core technical condition underlying permutation tests. We argue that working through data scenarios that go beyond two-samples or have a non-*iid* structure provides a foundation of exchangeability to further develop permutation tests and hypothesis testing in general. Our goal is to present the key ideas underlying permutation methods, explore additional complications, and motivate how and why an investigation into permutation-based inference is valuable as a way to expand understanding of inference and statistical foundations.

1.1 Exchangeability

At the foundation of valid permutation tests is the condition of *exchangeability* (Pitman 1937; Good 2002). While there are typically no distributional or sample size restrictions on a permutation test, recognizing how to incorporate the exchangeability condition can sometimes be difficult (Welch 1990).

We use the following definition of exchangeability throughout:

> Data are **exchangeable under the null hypothesis** if the joint distribution from which the data came is the same before a permutation as after a permutation when the null hypothesis is true.

Good (2002) describes how permutation inference holds under "exchangeability-preserving transforms, asymptotic exchangeability, partial exchangeability, and weak exchangeability." His work, however, is quite theoretical and beyond the scope of a typical undergraduate course. Draper et al. (1993) gives a detailed account of the issues of exchangeability, including more complicated settings, such as serial correlation. Instead of proving theoretical properties of different exchangeability conditions, we walk the reader through examples that highlight different forms of exchangeability and violations of exchangeability.

1.2 Motivating example

Consider an example whose goal is to model performance in college as a quantitative outcome variable (Y) based on whether the student took Advanced Placement (AP) courses in high

school (X_1, the dichotomous "treatment" or "group" variable) and the student's family income level (X_2, a potential confounder). We begin by first assessing the association between AP course taking and college performance using data from a sample. For the moment, we leave aside X_2, and we assume that the families are independent and that only one student is included from each family. This example is then equivalent to a comparison of two groups, students who took AP courses and those who did not.

2 Permutation tests and exchangeability

2.1 How to permute?

Carrying out a permutation procedure involves calculation of a statistic for each of the permuted samples. The observed statistic is then compared to the permuted null distribution where the association between Y and X_1 has been broken.

Many settings, including a two-sample comparison, can be modeled using linear regression. For linear regression models with a single predictor (simple linear regression, SLR), equivalent permutation inference results will ensue no matter whether the outcome variable (Y) or the predictor variable (X_1) is permuted.

Figure 1 displays an example where interest is in comparing the mean across two groups and the observations are independent. There are eleven subjects, each from a different family; five of the subjects are exposed ($X_1 = 1$, e.g., the exposure group takes an AP course) and six of

the subjects are control samples ($X_1 = 0$, e.g., no AP course). The outcome variable (Y, e.g., college performance) is measured for all subjects.

In the example, we can either permute the outcome Y or the group indicator X_1. We note that because we are permuting, there are always five exposed observations and six control observations, leading to $\binom{11}{6} = 462$ possible permutations (although typically, in this or larger sample sizes, we won't calculate all possible permutations and instead sample from the distribution). We fit a SLR for each permutation and use the resulting null distribution of the t-statistics (i.e., the standardized slope coefficients for X_1) to assess the significance of the treatment (grouping) variable.

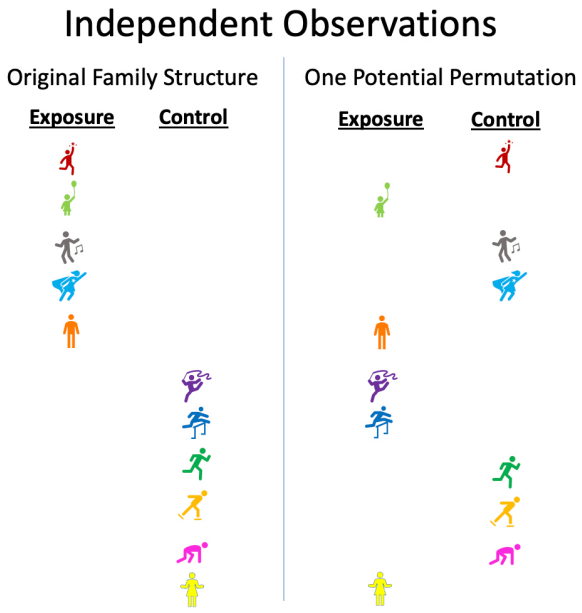


Figure 1: Each color and row represents a family structure. There are eleven individuals, none of whom are in the same family. Note that in the permutation setting, the exposure group and the control group both retain the same number of individuals as in the original family structure.

2.2 Clustered experimental design

Understanding exchangeability is sometimes best understood when it is violated. Consider a scenario where the underlying data are clustered in some fashion. For example, consider a situation where, for convenience or design purposes, families might be sampled rather than individuals. So instead of having just one individual from each family, we have two students from some families.

One of the conditions for inference in regression is that the observations (or more accurately, the residuals) are independent of one other. In the clustered data setting, it may no longer be tenable to assume that individuals within a family are independent of others within the same family.

Principled and flexible mathematical methods have been developed for a variety of models in the setting where the observations are clustered or repeated measures (see for example Laird and Ware (1982)). Random intercept models, where each observational unit (e.g., family) is assumed to have an underlying (unobserved) random level, are commonly used to account for clustering. However, such mixed effects models are both complicated (mixed effects models are not typically seen in undergraduate curricula) and may require a sufficient sample size to assume asymptotic normality of regression coefficients (Maas and Hox 2005).

Permutation methods are particularly desirable with clustered data since they can be adapted to more complicated settings where exchangeability might not be as straightforward as it is in the multiple linear regression (MLR) setting (Draper et al. 1993). In the next subsection, we

detail how violations of exchangeability in the clustered data setting can have a substantial impact on inferential results.

We revisit our motivating example where we are interested in predicting the outcome (college performance, Y) as a function of a dichotomous treatment (AP course taking, X_1) where some of the families have two individuals within the sample. An adaptation of the permutation tests that accounts for clustering within observational unit is a computational alternative to the large-sample (i.e., based on the Central Limit Theorem) random effects approach. The adapted permutation test is attractive because it maintains an appropriate Type I error rate (Winkler et al. 2014, 2015). Here we explore the clustered permutation idea with two different data scenarios, representing two different types of clustering that violate the independent observation condition.

2.2.1 Cluster: permuting homogeneous clustered observations

In the homogeneous clustered (i.e., all individuals within a family have the same value of the treatment variable) data scenario, there are multiple observations of the same type within each observational unit (see Figure 2). There are a total of eight families. Five of the families have a single observation, while three of the families have two individuals.

With homogeneous clustered observations, the permutation scheme must preserve the family structure determined by the experimental design. Here, each family has only one treatment (either exposure or control), so we permute the treatments with the constraint that three of the single-person families and one of the two-person families are in the control group. For

the homogeneous clustered example, there are $\binom{5}{3} \cdot \binom{3}{1} = 30$ possible permutations. We fit a regression for each permutation and use the resulting null distribution of the t-statistics (i.e., the standardized slope coefficients for X_1) to assess the significance of the treatment variable.

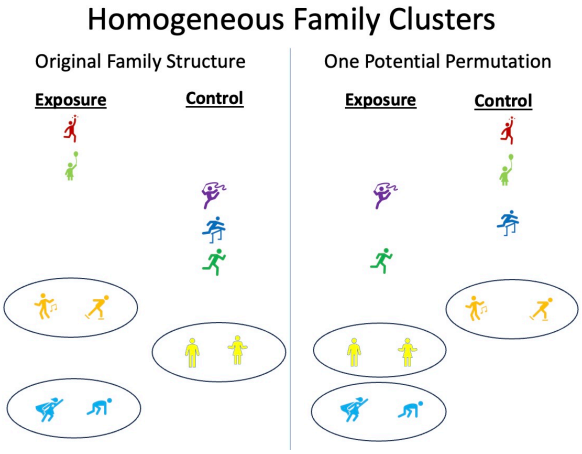


Figure 2: Each color and row represents a family structure. There are five single-person "families" and three families with two members each. In the original structure, treatment happens by family. Note that in the permutation setting, the exposure group and the control group both retain the same number of individuals as in the original family structure. The families are permuted together so as to keep intact the original scenario of treatment within family.

2.2.2 Cluster: permuting heterogeneous clustered observations

In the heterogeneous clustered (e.g., each family has members with both values of the dichotomous treatment) data scenario, families with two observations have individuals of each type of treatment observed within each observational unit (see Figure 3). There are a total of eight families. Five of the families have a single observation, and three of the families have two individuals.

With heterogeneous clustered observations, the permutation scheme must preserve the family structure. Here, the treatment on single-person families will be permuted as if the observations were independent (i.e., a permutation with the constraint that two individuals are exposed and three are control). The permutation of observations in the two-person families happens within the family. That is, with equal probability, the treatment levels are either kept the same or swapped, within each family.

For the heterogeneous clustered scenario, there are $\binom{5}{3} \cdot 2^3 = 80$ possible permutations. After each permutation, we fit a regression on the quantitative predictor and the permuted treatment. The resulting null distribution of the t-statistics is used to assess the significance of the treatment variable.

2.2.3 Simulation of clustered and independent scenarios

With the clustered family structure, if the analyst permutes either the group variable X_1 or the outcome Y naïvely (without regard to the family structure), there is a type of exchangeability violation due to the independence (or lack thereof) of observations. The independence exchangeability violation does matter (in terms of performance) as seen in what follows.

We undertook a limited series of simulation studies to explore the behavior of permutation tests when the design structure was intentionally specified to violate the independence condition. The simulations exploited the fact that ignoring clustering for the homogeneous clusters yields standard errors that are too small and that doing so for the heterogeneous cluster design,

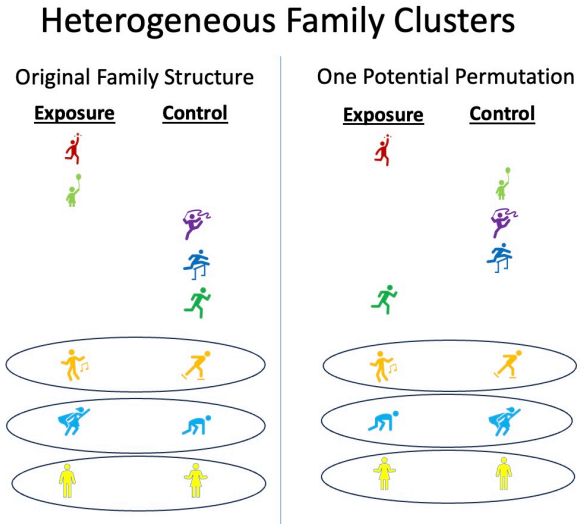


Figure 3: Each color and row represents a family structure. There are five single-person "families" and three families with two members each. In the original structure, for the families of size two, exactly one member of each family is exposed, and the other member of the family is in the control group. Note that in the permutation setting, the exposure group and the control group both end up with the same number of individuals as in the original family structure. For the families of size two, the permutation happens within the family so as to keep intact the original structure of both treatments across the family.

where family members serve as their own control, yields standard errors that are too large (Cannon et al. 2001).

The heart of our simulations is well described by Winkler et al. (2015), who focus on

> any dataset with known dependence among observations. In such cases, some permutations, if performed, would create data that would not possess the original dependence structure, and thus, should not be used to construct the reference (null) distribution. To allow permutation inference in such cases, we test the null hypothesis using only a subset of all otherwise possible permutations, i.e., using only the rearrangements of the data that respect exchangeability, thus retaining the original joint distribution unaltered.

For each model/scenario combination, 2000 simulations were generated with $n = 20$ subjects per treatment group (more subjects than displayed in Figure 1, Figure 2, and Figure 3). For the clustering scenarios, half of the subjects were singletons and the others shared a family member within the group (homogeneous) or across groups (heterogeneous). Within each simulation, the null distribution was created using 2000 permutations.

Table 1 displays the results from the simulation study. The `binom.test()` function in R was used to calculate a 95% CI for the estimated Type I error rates. As expected, the models that do not appropriately account for clustering are anti-conservative ("lm homogeneous", "naïve permutation homogeneous") or overly conservative ("lm heterogeneous" and "naïve permutation heterogeneous"); we have bolded the models that don't maintain the desired

Type I error rate of 0.05. Methods that appropriately account for the clustering (e.g., "lme" or "correct permutation") include the desired Type I error rate within the 95% CI. When there are cluster dependencies, the naïve permutation method is not advised.

Table 1: Type I error rates for different scenarios (2000 simulations, each with 2000 permutations for each simulation). Models include SLR (lm), linear mixed effects (lme), permutation not accounting for clustering (naïve), and permutation accounting for clustering (correct). Forms of clustering are independent (yellow), homogeneous (white), and heterogeneous (blue). The dichotomous treatment variable was permuted (Draper and Stoneman 1966). Type I error was computed using $\alpha = 0.05$.

Model/Scenario	Type I error rate (95% CI)	Results
lm independent	0.042 (0.034-0.052)	includes desired alpha level
correct permutation independent	0.046 (0.037-0.056)	includes desired alpha level
lm homogeneous	**0.07 (0.059-0.082)**	**anticonservative**
lme homogeneous	0.051 (0.041-0.061)	includes desired alpha level
naïve permutation homogeneous	**0.072 (0.061-0.084)**	**anticonservative**
correct permutation homogeneous	0.045 (0.036-0.055)	includes desired alpha level
lm heterogeneous	**0.029 (0.023-0.038)**	**conservative**
lme heterogeneous	0.044 (0.035-0.054)	includes desired alpha level
naïve permutation heterogeneous	**0.034 (0.027-0.043)**	**conservative**
correct permutation heterogeneous	0.056 (0.047-0.068)	includes desired alpha level

2.3 Multiple linear regression

It is well documented that work in AP courses is positively associated with socio-economic status (Kolluri 2018), which motivates the introduction of X_2 as a possible confounder of the relationship between X_1 and Y (Cummiskey et al. 2020). Throughout all of the four MLR hypothesis tests described below, assume that the null hypothesis ($\beta_1 = 0$) is true, that is, having taken an AP course (X_1) is not linearly related to performance in college (Y) after controlling for family income (X_2).

Unlike SLR, with MLR, there is no obvious choice of how to permute. In what follows, we discuss four different permutation methods and their exchangeability conditions. The goal of permuting is to create a null sampling distribution of the statistic of interest so that we may infer the variability of the statistic. That is, a distribution of the statistics under the setting where the outcome Y and the predictor of interest X_1 are not linearly related. But the null distribution also needs to be created under the exchangeability condition—the permuting only leads to a change in the relationship between Y and X_1 without changing any of the other variable relationships in the linear model.

Communicating the different permutation structures to students is an ideal way to explore the ideas of exchangeability. Even though the methods below are quite similar with respect to Type I errors and power, the *discussion* of the methods allows students to study exchangeability and permutations more generally. We include extended details of the permutation schemes in Appendix A.3. Here, we briefly describe the four permutation methods with specific thought

to the exchangeability violations for each one.

2.3.1 Permute Y

At first glance, it might seem like permuting the outcome variable would be a good way to break the relationship between Y and X_1 (Manly 1986, 1997) (see complete algorithm in Appendix A.3). Indeed, permuting Y will break the relationship between Y and X_1, which will force the null hypothesis to be true (which is what we want for testing). Additionally, permuting Y will preserve the relationship between X_1 and X_2 (as desired). However, permuting Y will also simultaneously break the relationship between Y and X_2, which may not be acceptable if we need to preserve the relationship to mirror the original data structure.

Breaking the relationship between Y and X_2 wouldn't be an issue if Y and X_2 were not associated. But if X_2 is a potential confounder of the relationship, then Y and X_2 may be associated. The permutation procedure where the outcome Y is permuted, therefore, is not exchangeable, even when the null hypothesis of interest is true. In this setting, permuting the outcome creates a permutation sampling distribution with the unintended consequence that performance in college (Y) is no longer associated with income level (X_2). Because performance and income **are** associated in the real world, the data with permuted Y values do not retain the observed dependency structure present in the original data. That is, a relevant relationship is lost after a permutation, which ends up violating the exchangeability condition.

The first block of rows in table Table 2 summarizes the broken and preserved relationships when permuting Y. Note that when Y is permuted, the original relationship between X_1 and

X_2 is preserved (which is what we want in terms of exchangeability). The broken relationship between X_2 and Y, however, is problematic in terms of exchangeability. If we permute Y, then exchangeability is met **only** if Y and X_2 are uncorrelated in the original dataset.

Table 2: Multiple linear regression models with two covariates; we show different permutation schemes, variable relationships that are broken, and variable relationships that are preserved. Any relationships that are broken, and **not** null, will violate exchangeability. Note that all methods require the observations to be independent of one another. See Appendix A for notation and full permutation schemes.

Permutation	Broken Relationships	Preserved Relationships	Permutation distribution
Permute Y	X_1 & Y	X_1 & X_2	$t^* = \frac{b^*_{1\cdot 2} - 0}{SE(b^*_{1\cdot 2})}$
Manly (1986, 1997)	X_2 & Y		See Eq (1)
Permute X_1	X_1 & X_2	X_2 & Y	$t^* = \frac{b^*_{1\cdot 2} - 0}{SE(b^*_{1\cdot 2})}$
Draper and Stoneman (1966)	X_1 & Y		See Eq (2)
Permute reduced model residuals	X_1 & Y (if X_1 & X_2 are uncorrelated)	X_1 & X_2	$t^* = \frac{b^*_{1\cdot 2} - 0}{SE(b^*_{1\cdot 2})}$
Freedman and Lane (1983)		X_2 & Y	See Eq (3)

Permutation	Broken Relationships	Preserved Relationships	Permutation distribution
Permute full model residuals	None	X_1 & X_2	$t^* = \frac{b^*_{1\cdot 2} - b_{1\cdot 2}}{SE(b^*_{1\cdot 2})}$
ter Braak (1990, 1992)		X_1 & Y X_2 & Y	See Eq (4)

2.3.2 Permute $\mathbf{X_1}$

In order to maintain the relationship between Y and X_2 (while still interested in the relationship between Y and X_1), we might consider permuting X_1 instead of Y (Draper and Stoneman 1966) (see complete algorithm in Appendix A.3). Indeed, the permutation distribution created from permuting X_1 will force the null hypothesis to be true. However, permuting X_1 has the side effect of breaking the relationship between X_1 and X_2 in the permuted data. If the data come from, for example, a randomized clinical trial (where X_1 is the treatment variable), then X_1 and X_2 will be independent in the original dataset, and permuting X_1 will on average **not** violate the exchangeability condition. If X_1 and X_2 are correlated in the original dataset, as seen in our AP course and socio-economic status example, then permuting X_1 violates the exchangeability condition. The second block of tows in Table 2 summarizes the broken and preserved relationships when permuting X_1. Note that when X_1 is permuted,

the original relationship between X_2 and Y is preserved (which is what we want in terms of exchangeability). The broken relationship between X_1 and X_2, however, is problematic under general conditions.

2.3.3 Permute reduced model residuals

In order to address some of the concerns of the first two MLR permutation methods, Freedman and Lane (1983) propose a permutation scheme based on residuals (see complete algorithm in Appendix A.3). The permutation preserves the relationship between X_1 & X_2 as well as the relationship between X_2 & Y. However, in order for the relationship between X_1 & Y (the key relationship!) to be broken, X_1 and X_2 must not be associated. The third block of rows in Table 2 summarizes the broken and preserved relationships when permuting the reduced model residuals. Note that when permuting the reduced model residuals, the original relationships between both X_1 & X_2 and additionally between X_2 & Y are preserved (which is what we want in terms of exchangeability). The null hypothesis is true only if the relationship between X_1 and Y is broken, and that happens only when X_1 and X_2 are uncorrelated. Appendix A.4 sketches the dependence of $\rho(Y^*, X_1)$ (the correlation between the permuted Y, Y^*, and X_1) on $\rho(X_1, X_2)$ (the correlation between X_1 and X_2).

2.3.4 Permute full model residuals

As an extension to Freedman and Lane (1983), ter Braak's (1990, 1992) permutation method permutes the residuals from the full model (see complete algorithm in Appendix A.3). The last

block of rows in Table 2 summarizes the broken and preserved relationships when permuting the full model residuals. Note that permuting the residuals under the full model allows all of the exchangeability conditions to hold. Unlike the previous three permutation schemes, here, the new model does not force the null hypothesis to be true, which is why the test statistic measures the deviation of the coefficients (on X_1) from the permuted data to the original data model coefficient (on X_1), instead of comparing to zero. The procedure works because the variability of the permuted coefficient around the sample coefficient mimics the variability of the sample coefficient around the parameter (ter Braak 1990). See Appendix A.3.4 for a more full justification of the use of the permutation statistic.

2.3.5 MLR: does it matter?

What impact does violation of exchangeability have on MLR inferences? We might consider studies of Type I error rate (when the null hypothesis is true) as well as Type II error rate (when the alternative hypothesis is true). Prior research (Anderson and Legendre 1999; Winkler et al. 2014) has shown that hypothesis testing in MLR is quite robust to the choice of permutation, and, except in cases of extreme error distributions, meeting (or not) the exchangeability conditions (such as breaking the association between Y and X_2) does not substantially impact the size or power of the test.

2.4 Comparison of correlation

A fourth example to help students think more deeply about exchangeability comes from a setting that does not have a parallel commonly taught standard mathematical test[2]. Consider the case where interest is in comparing the population correlation (ρ) across groups A ($X_1 = 1$) and B ($X_1 = 0$).

$$H_0 : \rho_A = \rho_B$$

$$H_A : \rho_A \neq \rho_B$$

The dataset includes three columns: one quantitative variable (X_2), the other quantitative variable (Y), and the group variable (X_1: Group A or Group B). A first glance at permuting would suggest to permute the group variable and use $r^*_A - r^*_B$ (the difference in sample correlations computed after permutation) as the test statistic given in the null sampling distribution. However, the permutation scheme violates exchangeability *unless* the models in A and B are the same in every way except the correlation.

Figure 4 illustrates ideas of both exchangeability and the null hypothesis. Figure 4.1.1 displays a sample from a world where the intercept, slope, and correlation are the same in groups A and B (i.e., the null hypothesis is true and all exchangeability conditions are met). Figure 4.1.2 displays the data and correlations in permuted groups. Because there is no difference

[2]The test of equality of correlations in two groups can be carried out using Fisher's r-to-z transformation (Fisher 1921; Diedenhofen and Musch 2015) but only with large samples (Stanley et al. 2024), and the procedure is not typically taught in an undergraduate curriculum.

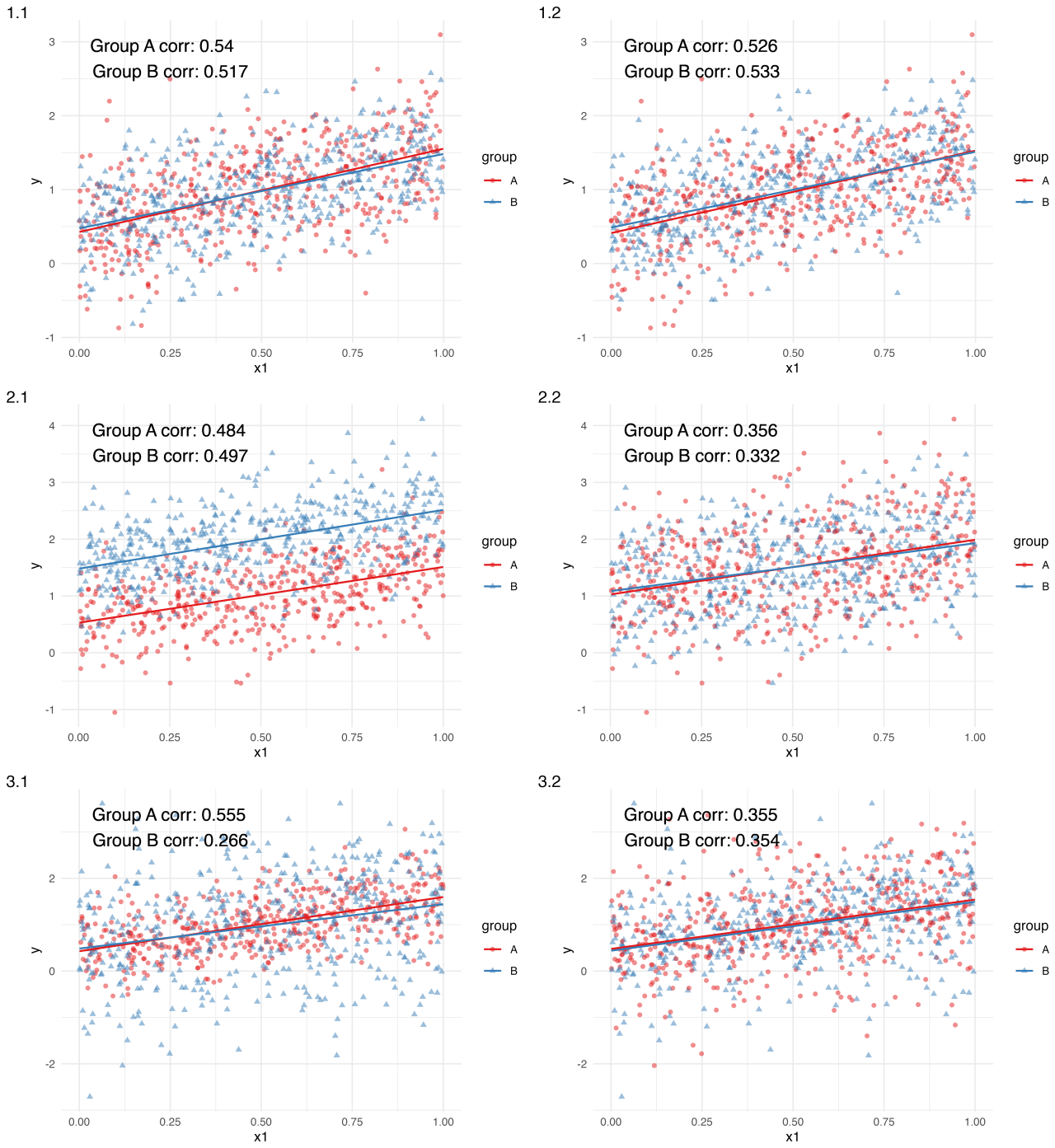


Figure 4: Results from 1. a setting where exchangeability holds and correlations are the same (Figure 4.1.1 displays sample data and Figure 4.1.2 displays a permuted sample); 2. a setting where the null hypothesis is true but exchangeability does not hold because the intercepts are different (Figue 4.2.1 displays sample data and Figure 4.2.2 displays a permuted sample with attenuated difference in correlations); and 3. a setting where exchangeability holds but correlations are different (4.3.1 displays sample data and 4.3.2 displays a permuted sample with attenuated difference in correlations).

between groups and exchangeability holds, the results from Figures 4.1.1 and 4.1.2 are quite similar.

However, Figures 4.2.1 and 4.2.2 highlight an arguably more common situation where the linear model across the two groups differs. In Figure 4.2.1, the null hypothesis happens to be true (the correlations are the same), but the violation of exchangeability (different intercepts) is important to notice regardless of whether or not the null hypothesis is true. Recall the definition of exchangeability which states that the data are exchangeable under the null hypothesis only if their distribution is the same before and after permutation. In Figure 4.2.1, the null hypothesis is true, and the distribution of the permuted data (Figure 4.2.2) are clearly not the same as the original data. If exchangeability is violated, then the process of permuting the group variable may not lead to a valid hypothesis test (regardless of whether or not the null hypothesis is true).

Contrast this with Figures 4.3.1 and 4.3.2, which represent data where the intercept and slope are the same but the correlations are different (i.e., the null hypothesis is not true but exchangeability holds). The estimates of the difference in correlations from the permutation in Figure 4.3.2 is smaller than the observed sample in Figure 4.3.1. Permuting the group variable would not violate exchangeability because the models are identical except for the difference in correlation (i.e., except for the condition of the null hypothesis).

We carried out a simulation study to assess the impact of violations of the exchangeability assumption in this setting. We ran 5,000 simulations with n = 20 observations in each of the two groups. The true regression model was defined as $\beta_0 = 0.5$, $\beta_1 = 1$ with root MSE =

0.5. Here, the underlying correlation is 0.5 in both groups (corresponding to Figures 4.1.1 and 4.1.2). The permutation test which shuffled the group indicator and calculated the difference in correlations effectively maintained the desired Type 1 error rate of 0.05 (average Type I error of 0.052, 95% CI ranged from 0.046 to 0.059).

We next considered the situation described in the second row of Figure 4 (Figures 4.2.1 and 4.2.2), where the correlations remain the same (0.5 in both groups), but the intercepts are different (0.5 in group A and 1.5 in group B). In this scenario, where exchangeability is violated, the permutation procedure fails to maintain an appropriate Type I error rate (average Type I error of 0.024, 95% CI ranged from 0.020 to 0.029). Notice that the correlations in Figure 4.2.2 are substantially smaller than the correlations in Figure 4.2.1 (as expected). The variability of the correlation is inversely related to the magnitude of the correlation. Which is to say that the permuted null sampling distribution (from permutations like those seen in Figure 4.2.2) will be wider than the theoretical null sampling distribution from data like those in Figure 4.2.1. Because the permuted null sampling distribution is wider, the Type I error rate is lower than the nominal 0.05 level.

For the bottom row of Figure 4 (Figures 4.3.1 and 4.3.2), we did not test Type I error because the null hypothesis is false. We could have done a power analysis, but assessing power under various settings was beyond the scope of the paper.

There are multiple other linear regression settings where students can consider exchangeability violations and hypothesis conditions. For example, an extension of Figure 4 would be to create a setting where the slopes differed and the alternative hypothesis was true (e.g., residuals are

bigger in one plot). Additionally, changing the range of the X_1 values across Group A and Group B would also violate exchangeability (as would changing the range of the Y values across Groups A and B).

3 Discussion and pedagogical recommendations

Sometimes the easiest way to understand an assumption is to explore when the assumption is violated. In this paper we provide four classes of examples and discuss what exchangeability means in each. We discuss how the analysis does or does not violate exchangeability, depending on the permutation scheme itself. We hope that the examples prompt students to ask themselves "is this even valid given the question of interest and properties of the data?" when running a permutation test.

We do not provide the examples to suggest that permutation tests are or are not superior to their mathematical counterparts. Indeed, as seen in the MLR setting, the literature is clear that some violations of exchangeability do not have a substantial impact on the Type I error.

Instead, we provide the examples in order to continue the conversation surrounding permutation tests at the undergraduate level. We hope that our examples will be used in the classroom as a way for teachers to explain to students that permutation tests **do** have technical conditions. Just like other tests in the introductory canon (e.g., the t-test), the permutation test is more robust to violations of some conditions and less robust to violations of others.

The sophisticated and extremely powerful concept of exchangeability can be accessed through

examples presented in the classroom. After digging into the exchangeability ideas, students are able to apply a permutation approach to more complicated data settings (which are often encountered in real applications). Our approach is consistent with the advice of Chance et al. (2024), who hoped "to help instructors understand the differences in the simulation strategies to better inform their own decisions of how to adapt a simulation approach for their classes and to better respond to student questions that may arise."

Another compelling rationale (from Vanhove (2015)) to devote time to these questions is the concept that "the test follows from the design":

> There is a clear logical link between the statistical test we use and the experimental design we opted for: using a permutation test is entirely warranted by the random assignment of participants to two equal-sized groups. Stressing the link between experimental design and statistical inference – rather than considering them separately – is of huge pedagogical, as well as practical, use, I believe (Vanhove 2015).

In randomized studies, the link between experimental design and permutation scheme should be made clear to students. Shuffling treatment labels models random assignment of a single variable, while the clustered schemes model restricted randomization. An understanding of the experimental design can bring the appropriate permutation scheme to light.

3.1 Classroom recommendations

Experience teaching exchangeability

In a recent upper division undergraduate statistics elective class, one of the authors presented some of the ideas described in the paper. In particular, she asked her students how permuting X_1 violates exchangeability in the MLR setting. The answer starts by acknowledging that when X_1 is permuted, the relationship between Y and X_1 is broken, which is necessary for H_0 to be true. However, the relationship between X_1 and X_2 is also broken, and if X_1 and X_2 were correlated in the original data, they should be correlated in the permuted data as well (so exchangeability is broken). The following represent paraphrases of the type of answers students gave to the in-class activity. The responses are illustrative anecdotes; they are not meant to indicate a proportional representation of the suite of potential answers. Student-paraphrased responses are given in quotes, while our reflections are given in italics. The prompt given to the students is:

> Your friend proposes to permute the variable X_1 in order to create a null sampling distribution for the coefficient on X_1. In what way does your friend's permutation scheme violate the exchangeability condition?

- "Exchangeability means that the data before and after the permutation comes from the same distribution… we are incorrectly assuming that $X_2 = 0$, i.e., that X_2 does not matter." *The student is able to state the definition of exchangeability, but they confuse the idea of the coefficient being zero ($H_0 : \beta_2 = 0$) and the variable being zero. The student also misses the important aspect of the relationship between X_1 and X_2.*
- "If we originally had correlation, we no longer have correlation." *The student understood that aspects of the relationships needed to be the same originally and in the permuted*

data, but they did not specify which variables should be correlated.

- "Ignoring the other covariates, breaking the conditional structure for the test." *The student conflates the conditional structure of the data itself with some aspect of the test.*
- "If X_1 and X_2 are correlated then permuting X_1 will break that correlation / relationship." *The student understands the issue of violating exchangeability but does not discuss the motivation of permuting X_1 to create the null sampling distribution.*
- "By permuting X_1, we break the Y and X_1 relationship, but now we have also broken the relationship between X_1 and X_2." *The student is able to succinctly identify the problem.*

We close with some practical guidance about how to bring exchangeability and permutation tests into the classroom. Underlying each of the classroom activities is the learning goal of students assessing whether the resampling scheme actually matches the question of interest. The suggestions are organized by type of classroom.

introductory undergraduate

- Teacher creates a simulation program (Rmd/qmd file or Shiny App) where students can change the error structure within a MLR analysis and discover that the desired Type I error rates are not achieved only in the case when the errors are particularly egregious.
- Use existing applets[3] to walk through permuting different variables or residuals for inferential analyses.

[3]e.g., the Two-variable ANOVA applet by Rossman/Chance https://www.rossmanchance.com/applets/2021/twoway/twoway.html

- Use a coin for an in-class permutation test activity on paired observations (e.g., see the "Matched Pairs Applet" described in Section 22.2 of Chance and Rossman (2021), with the corresponding URL of https://www.rossmanchance.com/applets/2021/matchedpairs/MatchedPairs.htm).

intermediate / advanced undergraduate

- Teach permutation tests for MLR. Talk to your students about the **choice** they will have to make (permute Y? X_1? residuals?) and that all modeling contains **choices**. Choices are not usually objective or unbiased, so an ability to defend their choice is what gives them power as a statistician.
- Describe the difference between creating a null sampling distribution (i.e., ensuring the data distribution is the same before and after permuting) and establishing that exchangeability holds (i.e., making sure that the Type I error rates will be accurate). Have students describe which aspect of the permutation does which job.
- Come up with scenarios (or use our cluster scenarios!) where the students can figure out the correct permutation schemes. See an example from JH's class that uses permutations to address a research question using a stratified two-sample test.[4] Or have the students explore how to develop a test for symmetry of a distribution (e.g., the null hypothesis is that the distribution is symmetric).
- Carry out a variety of permutation tests using a MLR model from an example dataset.

[4] https://st47s.com/Math154/Notes/permschp.html#macnell-teaching-evaluations-stratified-two-sample-t-test

Have students check the implementations or provide some of the implementations and have them carry out the others.

advanced undergraduate

- Ask your students what it means that "the test follows from the design." Use the cluster scenarios and apply the opposite permutation scheme. Have them compare the permutation scheme and the experimental design to figure out what makes the most **sense** in terms of the relationship of the permuting and the design.
- Ask students to read the paper by Helwig (2019) and explore the implications for the MLR setting via the `nptest` package in R (Helwig 2023). Have them explore how and why robust permutation tests (DiCiccio and Romano 2017) use alternative test statistics.
- Have students run a simulation study to see if the Type I error rates given by the exchangeability violation in Figure 4 vary substantially from the level of significance.

advanced undergraduate / beginning graduate

- Present some or all of the regression methods described in Appendix A (with more in Winkler et al. (2014)) and have students fill out a blank Table 2 (see Appendix A). Ask students to report which relationships are broken and which are preserved.
- Find permutation tests in the literature and have students describe how the permutation scheme upholds (or doesn't!) the exchangeability condition.

graduate

- Find theoretical work that proves exchangeability and demonstrate the important mapping of the theory to the applied problems (or maybe the exchangeability is not vital to the performance of the method?).

We suggest some ideas at particular levels, but many of the pedagogical ideas can be adjusted to be effective at a variety of levels. Whether to permute Y or X_1 can be taught as early as introductory statistics, where students have done some hypothesis testing and some multivariate modeling. Graduate students can think carefully about how to prove that conditions of exchangeability are met. The group of students who might most benefit from the ideas we've presented include upper level undergraduates and early graduate students who are focused on understanding which models are best in which settings and how to differentiate those settings.

4 Acknowledgements

We thank the reviewers and associate editor for many helpful suggestions that have improved the manuscript.

5 Disclosure statement

The authors report there are no competing interests to declare.

6 Reproducibility statement

The code used to undertake the simulation studies reported in the paper have been made available at https://osf.io/2avw6.

References

Anderson, Marti J., and Pierre Legendre. 1999. "An Empirical Comparison of Permutation Methods for Tests of Partial Regression Coefficients in a Linear Model." *Journal of Statistical Computation and Simulation* 62 (3): 271–303. https://doi.org/10.1080/00949659908811936.

Baumer, Benjamin S., Daniel T. Kaplan, and Nicholas J. Horton. 2024. *Modern Data Science with R*. 3rd ed. Boca Raton, FL: CRC Press. https://mdsr-book.github.io/mdsr3e/.

Cannon, Michael J., Lee Warner, J. Augusto Taddei, and David G. Kleinbaum. 2001. "What Can Go Wrong When You Assume That Correlated Data Are Independent: An Illustration from the Evaluation of a Childhood Health Intervention in Brazil." *Statistics in Medicine* 20 (9-10): 1461–67. https://doi.org/10.1002/sim.682.

Çetinkaya-Rundel, M., and J. Hardin. 2023. *Introduction to Modern Statistics*. 2nd ed. https://openintro-ims.netlify.app/.

Chance, Beth, Karen McGaughey, Sophia Chung, Alex Goodman, Soma Roy, and Nathan Tintle. 2024. "Simulation-Based Inference: Random Sampling vs. Random Assignment? What Instructors Should Know." *Journal of Statistics and Data Science Education* 0 (0): 1–10. https://doi.org/10.1080/26939169.2024.2333736.

Chance, Beth, and Allan Rossman. 2021. *Investigating Statistical Concepts, Applications, and Methods*. 4th ed. https://www.rossmanchance.com/iscam3/.

Cox, D. R., and D. V. Hinkley. 1974. *Theoretical Statistics*. Chapman & Hall.

Cummiskey, Kevin, Bryan Adams, James Pleuss, Dusty Turner, Nicholas Clark, and Krista Watts. 2020. "Causal Inference in Introductory Statistics Courses." *Journal of Statistics Education* 28 (1): 2–8. https://doi.org/10.1080/10691898.2020.1713936.

DiCiccio, Cyrus J., and Joseph P. Romano. 2017. "Robust Permutation Tests For Correlation And Regression Coefficients." *Journal of the American Statistical Association* 112 (519): 1211–20. https://doi.org/10.1080/01621459.2016.1202117.

Diedenhofen, Birk, and Jochen Musch. 2015. "Cocor: A Comprehensive Solution for the Statistical Comparison of Correlations." *PLoS ONE* 10 (4). https://doi.org/10.1371/journal.pone.0121945.

Draper, David, James Hodges, Colin Mallows, and Daryl Pregibon. 1993. "Exchangeability and Data Analysis." *Journal of the Royal Statistical Society Series A: Statistics in Society* 156 (1): 9–28. https://doi.org/10.2307/2982858.

Draper, Norman, and David M. Stoneman. 1966. "Testing for the Inclusion of Variables in Linear Regression by a Randomisation Technique." *Technometrics* 8 (4): 695–99. http://www.jstor.org/stable/1266641.

Efron, Bradley. 1982. *The Jackknife, the Bootstrap and Other Resampling Plans*. Society for Industrial; Applied Mathematics.

Efron, Bradley, and Robert J. Tibshirani. 1994. *An Introduction to the Bootstrap*. 1st ed. Chapman; Hall/CRC. https://doi.org/https://doi.org/10.1201/9780429246593.

Fisher, R. A. 1935. *The Design of Experiments*. New York: Hafner.

Fisher, Ronald A. 1921. "On the "probable error" of a Coefficient of Correlation Deduced from a Small Sample." *Metron* 1: 3–32.

Freedman, David, and David Lane. 1983. "A Nonstochastic Interpretation of Reported Significance Levels." *Journal of Business & Economic Statistics* 1 (4): 292–98. https://doi.org/10.2307/1391660.

Garfield, Joan, Robert delMas, and Andrew Zieffler. 2012. "Developing Statistical Modelers and Thinkers in an Introductory, Tertiary-Level Statistics Course." *ZDM Mathematics Education* 44 (7): 883–98. https://doi.org/10.1007/s11858-012-0447-5.

Good, P. 2002. "Extensions of the Concept of Exchangeability and Their Applications." *Journal of Modern Applied Statistical Methods* 1 (2): 243–47. https://doi.org/10.22237/jmasm/1036110240.

Helwig, Nathaniel E. 2019. "Robust Nonparametric Tests of General Linear Model Coefficients: A Comparison of Permutation Methods and Test Statistics." *NeuroImage* 201: 116030. https://doi.org/https://doi.org/10.1016/j.neuroimage.2019.116030.

Helwig, Nathaniel E. 2023. *Nptest: Nonparametric Bootstrap and Permutation Tests*. https://doi.org/10.32614/CRAN.package.nptest.

Hesterberg, Tim C. 2015. "What Teachers Should Know about the Bootstrap: Resampling in the Undergraduate Statistics Curriculum." *The American Statistician* 69 (4): 371–86. https://doi.org/10.1080/00031305.2015.1089789.

Ismay, Chester, and Albert Kim. 2020. *ModernDive: Statistical Inference via Data Science*. Boca Raton, FL: CRC Press. https://moderndive.com/.

Kolluri, Suneal. 2018. "Advanced Placement: The Dual Challenge of Equal Access and Effectiveness." *Review of Educational Research* 88: 671–711. https://journals.sagepub.com/doi/10.3102/0034654318787268.

Laird, N M, and J H Ware. 1982. "Random-Effects Models for Longitudinal Data." *Biometrics* 38: 963–74.

Lehmann, E. L., and H. J. M D'Abrera. 1975. *Nonparametrics: Statistical Methods Based on Ranks*. Holden-Day, San Francisco.

Lock, Robin H., Patti Frazer Lock, Kari Lock Morgan, Eric F. Lock, and Dennis F. Lock. 2025. *Statistics: Unlocking the Power of Data*. 4th ed. John Wiley & Sons. https://www.wiley.com/en-us/Statistics%3A+Unlocking+the+Power+of+Data%2C+4th+Edition-p-9781394336579.

Maas, C J M, and J J Hox. 2005. "Sufficient Sample Sizes for Multilevel Modeling." *Methodology* 1: 86–92. https://doi.org/10.1027/1614-2241.1.3.86.

Manly, Bryan F. J. 1986. "Randomization and Regression Methods for Testing for Associations with Geographical, Environmental and Biological Distances Between Populations." *Population Ecology* 28 (2): 201–18. https://doi.org/https://doi.org/10.1007/BF02515450.

Manly, Bryan F. J. 1997. *Randomization, Bootstrap and Monte Carlo Methods in Biology*. 2nd ed. Texts in Statistical Science Series. London, UK: Chapman & Hall. https://doi.org/10.1080/02331887708801385.

Pitman, E. J. G. 1937. "Significance Tests Which May Be Applied to Samples from Any Populations." *Supplement to the Journal of the Royal Statistical Society* 4 (1): 119–30. http://www.jstor.org/stable/2984124.

Slutsky, E. 1925. "Über Stochastische Asymptoten Und Grenzwerte." *Metron* 5: 3–89.

Stanley, T. D., Hristos Doucouliagos, Maximilian Maier, and František Bartoš. 2024. "Correcting Bias in the Meta-Analysis of Correlations." *Psychological Methods*, ahead of print. https://doi.org/10.1037/met0000662.

Tantawanich, Siriwan. 2006. "Permutation Test for a Multiple Linear Regression Model." https://api.semanticscholar.org/CorpusID:201824760.

Taylor, Laura, and Kirsten Doehler. 2015. "Reinforcing Sampling Distributions Through a Randomization-Based Activity for Introducing ANOVA." *Journal of Statistics Education* 23 (3): 1–33. https://jse.amstat.org/v23n3/taylor.pdf.

ter Braak, C. J. F. 1990. "Update Notes: CANOCO, Version 3.10." *Wageningen, The Nether-*

lands: Agricultural Mathematics Group, January, 35.

ter Braak, C. J. F. 1992. "Permutation Versus Bootstrap Significance Tests in Multiple Regression and Anova." *Austr. J. Statist.* 29 (January): 79–85. https://doi.org/10.1007/978-3-642-48850-4_10.

Vanhove, Jan. 2015. *Explaining Key Concepts Using Permutation Tests*. https://janhove.github.io/posts/2015-02-26-explaining-key-concepts-using-permutation-tests.

Welch, W. 1990. "Construction of Permutation Tests." *Journal of the American Statistical Association* 85: 693–98.

Winkler, A. M., G. R. Ridgway, M. A. Webster, S. M. Smith, and T. E. Nichols. 2014. "Permutation Inference for the General Linear Model." *NeuroImage* 92: 381–97.

Winkler, A. M., M. A. Webster, D. Vidaurre, T. E. Nichols, and S. M. Smith. 2015. "Multi-level Block Permutation." *NeuroImage* 123: 253–68. https://doi.org/10.1016/j.neuroimage.2015.05.092.

Ye, Julie. 2023. "Permutation Tests for Multiple Linear Regression Models." https://hardin47.github.io/st47s-and-d47a/student-work/julie_ye_2023.pdf.

A Appendix

Here we review four different approaches for carrying out permutation tests for MLR models. We provide some theoretical content that might be of interest to an advanced undergraduate or early graduate student.

A.1 Why permutation tests?

As previously observed, even when both the normality conditions and the exchangeability are modestly violated, all four of the permutation test methods described below do reasonably well at controlling Type I errors (Anderson and Legendre 1999; Winkler et al. 2014). When error terms come from extremely skewed distributions (e.g., cubed exponentials), the permutation tests maintained the appropriate Type I error rate while normal theory methods were overly conservative (Anderson and Legendre 1999). There is strong evidence that in order to violate alpha-level testing, the data must violate the technical conditions to an extreme degree (Anderson and Legendre 1999; Tantawanich 2006; Winkler et al. 2014).

If we anticipate that results will be indistinguishable, why is an understanding of permutation methods (and exchangeability) important? Why do we present the methods below as valuable for communicating exchangeability to our students?

1. We believe that exchangeability is a valuable concept for students to understand, as it undergirds key foundational knowledge of statistical inference.

2. There do exist settings where mathematical (i.e., based on the Central Limit Theorem) and permutation approaches produce different inferential results, even within the linear model framework. Knowledge of permutation-based approaches (plus computational skills) allows inference in such areas where linear model conditions don't hold, and permutation tests may be straightforward to implement in situations where typical parametric tests may exist but be extremely complicated (e.g., dependence in the experimental design, as described in Section 2.2). As long as permuted draws can be made from the appropriate null world, permutation tests can be used to make inferences without parametric conditions.

A.2 Set-up

In order to work through the details of the permutation methods and corresponding exchangeability constraints, we provide notation and model specifications to describe the MLR model.

The population model of interest is

$$E[Y|X_1, X_2] = \beta_{0\cdot 1,2} + \beta_{1\cdot 2}X_1 + \beta_{2\cdot 1}X_2,$$

where X_1 is the variable of interest in predicting Y, and X_2 is a nuisance variable. The $\cdot$ (dot) notation indicates the other predictor variables included in the model.

The model estimated from the original dataset is given by

$$\widehat{Y} = b_{0\cdot 1,2} + b_{1\cdot 2}X_1 + b_{2\cdot 1}X_2,$$

where $b_{1\cdot 2}$ is the sample coefficient on X_1 given that X_2 is in the model, and $b_{2\cdot 1}$ is the sample coefficient on X_2 given that X_1 is in the model.

An equivalent framing to the original data model is given for the permuted data. For example, if Y is permuted to get Y^*, then the model estimated from the permuted dataset is given by

$$\widehat{Y^*} = b^*_{0\cdot 1,2} + b^*_{1\cdot 2}X_1 + b^*_{2\cdot 1}X_2,$$

where $b^*_{1\cdot 2}$ is the sample coefficient on X_1 given that X_2 is in the model, and $b^*_{2\cdot 1}$ is the sample coefficient on X_2 given that X_1 is in the model, while it is Y^* being regressed on X_1 and X_2.

All standard errors of the coefficients ($SE(b)$ and $SE(b^*)$) are calculated using the normal ordinary least squares (OLS) formula used in standard linear regression software. Additionally, it is worth pointing out that underlying all of the methods presented in Appendix A are observations, which are assumed to be independent of one another.

A.3 Permutation algorithms

A.3.1 Permute Y

Summary of algorithm when permuting Y (Manly 1986)

1. Fit the original model and obtain coefficient estimates ($b_{0\cdot1,2}$, $b_{1\cdot2}$, and $b_{2\cdot1}$) and corresponding standard error estimates ($SE(b_{0\cdot1,2})$, $SE(b_{1\cdot2})$, and $SE(b_{2\cdot1})$):

$$\widehat{Y} = b_{0\cdot1,2} + b_{1\cdot2}X_1 + b_{2\cdot1}X_2$$

2. Permute Y to obtain Y^*.

3. Fit a model, now including X_1, on the permuted Y^* values to obtain permuted coefficient estimates ($b^*_{0\cdot1,2}$, $b^*_{1\cdot2}$, and $b^*_{2\cdot1}$) and corresponding standard error estimates ($SE(b^*_{0\cdot1,2})$, $SE(b^*_{1\cdot2})$, and $SE(b^*_{2\cdot1})$):

$$\widehat{Y^*} = b^*_{0\cdot1,2} + b^*_{1\cdot2}X_1 + b^*_{2\cdot1}X_2$$

4. Repeat steps 2 and 3 P times. For example, $P = 1000$.

5. From the P copies of $b^*_{1\cdot2}$ and P copies of $SE(b^*_{1\cdot2})$, calculate P copies of t^* to form the permuted null sampling distribution:

$$t^* = \frac{b^*_{1\cdot2} - 0}{SE(b^*_{1\cdot2})} \tag{1}$$

6. Compare the observed test statistic to the permuted null sampling distribution from step 5:

$$t_{obs} = \frac{b_{1\cdot2} - 0}{SE(b_{1\cdot2})}$$

A.3.2 Permute X_1

Summary of algorithm when permuting X_1 (Draper and Stoneman 1966)

1. Fit the original model and obtain coefficient estimates and corresponding standard error estimates:

$$\widehat{Y} = b_{0\cdot 1,2} + b_{1\cdot 2}X_1 + b_{2\cdot 1}X_2$$

2. Permute X_1 to obtain X_1^*.

3. Fit a model on the permuted X_1^* values to obtain permuted coefficient estimates and corresponding standard error estimates:

$$\widehat{Y} = b^*_{0\cdot 1,2} + b^*_{1\cdot 2}X_1^* + b^*_{2\cdot 1}X_2$$

4. Repeat steps 2 and 3 P times.

5. From the P copies of $b^*_{1\cdot 2}$ and P copies of $SE(b^*_{1\cdot 2})$, calculate P copies of t^* to form the permuted null sampling distribution:

$$t^* = \frac{b^*_{1\cdot 2} - 0}{SE(b^*_{1\cdot 2})} \tag{2}$$

6. Compare the observed test statistic to the permuted null sampling distribution from step 5:

$$t_{obs} = \frac{b_{1\cdot 2} - 0}{SE(b_{1\cdot 2})}$$

A.3.3 Permute reduced model residuals

Summary of algorithm when permuting reduced model residuals (Freedman and Lane 1983)

1. Fit the original model on X_2 only and obtain coefficient estimates and corresponding standard error estimates of the reduced model:

$$\widehat{Y} = b_{0\cdot 2} + b_2 X_2$$

2. Let the residuals be defined by $R_{Y\cdot 2} = Y - b_{0\cdot 2} - b_2 X_2$, and permute $R_{Y\cdot 2}$ to obtain $R^*_{Y\cdot 2}$. Define the permuted outcome variable as $Y^* = b_{0\cdot 2} + b_2 X_2 + R^*_{Y\cdot 2}$.

3. Fit a model on the permuted Y^* values to obtain permuted coefficient estimates and corresponding standard error estimates:

$$\widehat{Y^*} = b^*_{0\cdot 1,2} + b^*_{1\cdot 2} X_1 + b^*_{2\cdot 1} X_2$$

4. Repeat steps 2 and 3 P times.

5. From the P copies of $b^*_{1\cdot 2}$ and P copies of $SE(b^*_{1\cdot 2})$, calculate P copies of t^* to form the permuted null sampling distribution:

$$t^* = \frac{b^*_{1\cdot 2} - 0}{SE(b^*_{1\cdot 2})} \tag{3}$$

6. Compare the observed test statistic to the permuted null sampling distribution from step 5:

$$t_{obs} = \frac{b_{1\cdot 2} - 0}{SE(b_{1\cdot 2})}$$

It is not immediately obvious that a high correlation between X_1 and X_2 leads to a non-null distribution. We provide evidence for the claim in two ways: by describing the permutation scheme and by giving a sketch for the proof of the claim.

Claim. The correlation between Y^* and X_1 is dependent on the correlation between X_1 and X_2.

(1) The permutation scheme is designed to create new values for the coefficient on X_1 under the setting that X_1 and Y are uncorrelated given X_2. Repeated sets of the "permuted" Y^* values are generated by adding noise (permuted residuals) to the fitted values from the model on X_2 only. At first glance, it seems as though the method creates Y^* values that are associated with X_2 (because they built from the X_2-model fitted values) and not associated with X_1 (because they are built from a model that completely ignores X_1). However, if X_1 and X_2 are correlated, then Y^* values that are correlated with X_2 will naturally be correlated with X_1.

(2) In Appendix A.4, we give a proof sketch outlining the derivation for the covariance between Y^* and X_1. Although the proof is only outlined, the dependence on the correlation between X_1 and X_2 is clear. The approximate value of $\rho(Y^*, X_1)$ derived in Appendix A.4 is validated in Ye (2023).

A.3.4 Permute full model residuals

Summary of algorithm when permuting full model residuals (ter Braak 1990)

1. Fit the original model and obtain coefficient estimates and corresponding standard error estimates of the full model:

$$\widehat{Y} = b_{0 \cdot 1,2} + b_{1 \cdot 2} X_1 + b_{2 \cdot 1} X_2$$

2. Let the residuals $R_{Y \cdot 1,2} = Y - b_{0 \cdot 1,2} - b_{1 \cdot 2} X_1 - b_{2 \cdot 1} X_2$, and permute $R_{Y \cdot 1,2}$ to obtain $R^*_{Y \cdot 1,2}$. Define the permuted outcome variable as $Y^* = b_{0 \cdot 1,2} + b_{1 \cdot 2} X_1 + b_{2 \cdot 1} X_2 + R^*_{Y \cdot 1,2}$.

3. Fit a model on the permuted Y^* values to obtain permuted coefficient estimates and corresponding standard error estimates:

$$\widehat{Y^*} = b^*_{0 \cdot 1,2} + b^*_{1 \cdot 2} X_1 + b^*_{2 \cdot 1} X_2$$

4. Repeat steps 2 and 3 P times.

5. From the P copies of $b^*_{1 \cdot 2}$ and P copies of $SE(b^*_{1 \cdot 2})$, calculate P copies of t^* to form the permuted null sampling distribution:

$$t^* = \frac{b^*_{1 \cdot 2} - b_{1 \cdot 2}}{SE(b^*_{1 \cdot 2})} \tag{4}$$

6. Compare the observed test statistic to the permuted null sampling distribution from step 5:

$$t_{obs} = \frac{b_{1\cdot2} - 0}{SE(b_{1\cdot2})}$$

Note that the goal here is to understand the sampling distribution of the statistic $b^*_{1\cdot2}$ without specifying a null hypothesis–exactly the work of bootstrapping (caveat: the work here is done without replacement, where most bootstrapping is done with replacement). Similar to the development of other bootstrapping methods (Efron and Tibshirani 1994, 87), the underlying technical condition for ter Braak's method is that:

$$F(x)_{\left(\frac{b^*_{1\cdot2} - b_{1\cdot2}}{SE(b^*_{1\cdot2})}\right)} \approx F(x)_{\left(\frac{b_{1\cdot2} - \beta_{1\cdot2}}{SE(b_{1\cdot2})}\right)}, \tag{5}$$

where F is the cumulative distribution function of the random variable described in the subscript of the function.

The sampling distribution of $t^* = \frac{b^*_{1\cdot2} - b_{1\cdot2}}{SE(b^*_{1\cdot2})}$ approximates the sampling distribution of $t = \frac{b_{1\cdot2} - \beta_{1\cdot2}}{SE(b_{1\cdot2})}$ due to the theoretical underpinnings from bootstrapping. That is, the variability of $b^*_{1\cdot2}$ around $b_{1\cdot2}$ mimics the variability of $b_{1\cdot2}$ around $\beta_{1\cdot2}$.

Under Equation (5), we can use the t^* distribution constructed from many permutations of the same dataset to carry out a hypothesis test. If H_0: $\beta_{1\cdot2} = 0$ is true, then $t_{obs} = \frac{b_{1\cdot2} - 0}{SE(b_{1\cdot2})}$ would be a likely value in the t^* distribution, corresponding to a non-significant p-value, resulting in a failure to reject H_0.

However, if H_A: $\beta_{1\cdot 2} \neq 0$ is true and, say, $\beta_{1\cdot 2} = 47$, then we would expect $\hat{t} = \frac{b_{1\cdot 2} - 47}{SE(b_{1\cdot 2})}$ to lie well within the t^* distribution, while $t_{obs} = \frac{b_{1\cdot 2} - 0}{SE(b_{1\cdot 2})}$ would lie in the margins, leading to a small p-value that concludes the test by rejecting H_0.

Furthermore, ter Braak (1990) synthesizes equations from various authors – Efron (1982) for the bootstrap, along with Cox and Hinkley (1974) and Lehmann and D'Abrera (1975) for the permutation – to make the following statements regarding the expected value and variance of the estimated slope coefficients in the bootstrapping versus permutation settings. $b^+_{1\cdot 2}$ corresponds to the coefficient derived from a standard bootstrap (full model), while $b^*_{1\cdot 2}$ corresponds to the permutation of the residuals under the full model.

$$E^+(b^+_{1\cdot 2}) = E^*(b^*_{1\cdot 2}) = b_{1\cdot 2} \tag{6}$$

$$var^+(b^+_{1\cdot 2}) = (1 - 1/n) var^*(b^*_{1\cdot 2}). \tag{7}$$

Equation (6) indicates that the expected values of $b^+_{1\cdot 2}$ and $b^*_{1\cdot 2}$ are both $b_{1\cdot 2}$. Equation (7) suggests that the variance of $b^+_{1\cdot 2}$, the bootstrapped estimate, is smaller than the variance of $b^*_{1\cdot 2}$. Hence, ter Braak uses Equations (6) and (7), along with the order property that $b^+_{1\cdot 2}$ and $b^*_{1\cdot 2}$ differ by $O(1/n)$ in second or higher order moments, to justify his proposal of the full model residual permutation strategy.

A.4 Impact of correlation between X_1 and X_2

Claim. The correlation between Y^*, derived from permuting resdiuals from the reduced model, and X_1 is dependent on the correlation between X_1 and X_2.

Proof sketch. Throughout the proof sketch, there are places where we have simplified the argument by considering statistics to be fixed parameters (not an unreasonable approximation under large sample sizes where Slutsky's Theorem holds (Slutsky 1925)).

$$\rho(Y^*, X_1) = \frac{cov(Y^*, X_1)}{\sqrt{var(Y^*)var(X_1)}}$$

Breaking down each part of the correlation between Y^* and X_1, we compute both $cov(Y^*, X_1)$ and $var(Y^*)$.

$$\begin{aligned} cov(Y^*, X_1) &= cov(b_{0\cdot 2} + b_2 X_2 + R^*_{Y\cdot 2}, X_1) \\ &= cov(b_{0\cdot 2}, X_1) + cov(b_2 X_2, X_1) + cov(R^*_{Y\cdot 2}, X_1) \\ &\approx cov(b_2 X_2, X_1) \qquad (8) \\ &\approx b_2 \cdot cov(X_1, X_2) \qquad (9) \\ &= b_2 \cdot \rho(X_1, X_2)\sqrt{var(X_1)var(X_2)} \end{aligned}$$

Note that $b_{0\cdot 2}$ and b_2 are random variables because they are statistics, so Equations (9,10,11) are all approximate. In Equation (8), we assume that $b_{0\cdot 2}$ and $R^*_{Y\cdot 2}$ are both independent of

X_1, which leads to $cov(b_{0\cdot 2}, X_1) = cov(R^*_{Y\cdot 2}, X_1) = 0$.

$$\begin{aligned} var(Y^*) &= var(b_{0\cdot 2} + b_2 X_2 + R^*_{Y\cdot 2}) \\ &\approx (b_2)^2 var(X_2) + var(R^*_{Y\cdot 2}) \end{aligned} \tag{10}$$

$var(R^*_{Y\cdot 2})$ is broken down into pieces.

$$\begin{aligned} var(R^*_{Y\cdot 2}) &= var(R_{Y\cdot 2}) \\ &= var(Y - b_{0\cdot 2} - b_2 X_2) \\ &\approx var(Y) + (b_2)^2 var(X_2) - 2b_2 cov(Y, X_2) \end{aligned} \tag{11}$$

$$\begin{aligned} cov(Y, X_2) &= cov(\beta_{0\cdot 1,2} + \beta_{1\cdot 2} X_1 + \beta_{2\cdot 1} X_2 + \varepsilon, X_2) \\ &= cov(\beta_{0\cdot 1,2}, X_2) + cov(\beta_{1\cdot 2} X_1, X_2) \\ &\quad + cov(\beta_{2\cdot 1} X_2, X_2) + cov(\varepsilon, X_2) \\ &= \beta_{2\cdot 1} var(X_2) + \beta_{1\cdot 2} cov(X_1, X_2) \end{aligned} \tag{12}$$

Plugging $cov(Y, X_2)$ into $var(R^*_{Y\cdot 2})$ and $var(R^*_{Y\cdot 2})$ into $var(Y^*)$, we can approximate

$var(Y^*)$.

$$
\begin{aligned}
var(Y^*) &\approx (b_2)^2 var(X_2) + var(Y) + (b_2)^2 var(X_2) \\
&\quad - 2b_2[\beta_{2\cdot1} var(X_2) + \beta_{1\cdot2} cov(X_1, X_2)] \\
&= 2(b_2)^2 var(X_2) + var(Y) \\
&\quad - 2b_2[\beta_{2\cdot1} var(X_2) + \beta_{1\cdot2} cov(X_1, X_2)] \\
&= 2b_2(b_2 - \beta_{2\cdot1}) var(X_2) + var(Y) - 2b_2 \beta_{1\cdot2} cov(X_1, X_2) \\
&= 2b_2[(b_2 - \beta_{2\cdot1}) var(X_2) - \beta_{1\cdot2} cov(X_1, X_2)] + var(Y)
\end{aligned}
$$

Putting it all together gives:

$$
\begin{aligned}
\rho(Y^*, X_1) &= \frac{cov(Y^*, X_1)}{\sqrt{var(Y^*) var(X_1)}} \\
&\approx \frac{b_2 \cdot \rho(X_1, X_2) \sqrt{var(X_1) var(X_2)}}{\sqrt{var(Y^*) var(X_1)}} \\
&= \frac{b_2 \cdot \rho(X_1, X_2) \sqrt{var(X_2)}}{\sqrt{var(Y^*)}} \\
&= \frac{b_2 \cdot \rho(X_1, X_2) \sqrt{var(X_2)}}{\sqrt{2b_2[(b_2 - \beta_{2\cdot1}) var(X_2) - \beta_{1\cdot2} cov(X_1, X_2)] + var(Y)}}. \qquad (13)
\end{aligned}
$$